The Essence of Civilization: A Formal Theory of Reciprocal Relations, Justice, and Conflict

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14 October 2026

Abstract

This paper develops a formal theory of civilization as an institutionalized order of human relations. It advances three claims. First, every relation can be classified by the sign and sustainability of its net transfers: relations are *symbiotic* when both parties reproduce their capacity, *parasitic* when one party extracts from a host that survives, and *predatory* when extraction destroys the host's capacity to reproduce. Second, stable civilizations require a justified equilibrium between contributions and benefits; when this equilibrium is violated beyond a threshold, conflict follows. Third, justice is the institutional mechanism that rebalances the ledger. The paper distinguishes civilization from its products: art, architecture, law, science, mathematics, and engineering are outcomes of the relational order, not the order itself, and their presence is no evidence of justice. The paper formalizes these claims using cooperative and non-cooperative game theory, principal–agent models, optimal extraction dynamics, evolutionary game theory, and dynamical systems. It derives the conditions for symbiotic viability, the patience threshold for sustainable cooperation, the extraction ceiling imposed by exit and voice, the conservation principle governing predation, the legitimacy constraint on extraction, the institutional conditions for cooperation as an evolutionarily stable strategy, and the phase transitions that generate collapse and renewal. The result is a unified theoretical apparatus for the comparative study of civilizations.

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1 Introduction

The problem of order is the foundational problem of social theory. Why do large-scale human orders cohere? Why do they break down? Answers have been given in terms of power, religion, law, culture, and economic structure. This paper proposes a more primitive principle: civilization is an ongoing negotiation over human relations—who gives, who gains, who loses, and how those outcomes are justified. A preliminary clarification is necessary. The word "civilization" is commonly attached to its products: monuments, codes, treatises, technologies. This paper treats those products as *outcomes* of the relational order, not as the order itself. Section 3 formalizes this distinction and shows that achievement is neither necessary nor sufficient for justice. A civilization may build pyramids and enslave their builders; it may codify law and exempt its elites; it may advance mathematics and suppress its heretics. The analytical object is the ledger, not the monument. The theory rests on a single observation. Every durable human order must solve a *reciprocity problem*: it must align contributions with benefits closely enough that participants reproduce their capacity to participate. When alignment fails, relations become extractive; when extraction exceeds the host's capacity to sustain it, relations become destructive; when destruction proceeds far enough, the order collapses. This observation yields three formal objects: a *classification* of relations, a *threshold* separating stability from instability, and a *mechanism*—justice—that restores the balance. The paper builds these objects axiomatically and derives their consequences. The argument proceeds as follows. Section 2 establishes primitives. Section 3 distinguishes the civilizational process from its products. Section 4 classifies relations. Section 5 derives the viability set. Section 6 analyzes repetition and the sustainability of cooperation. Section 7 models extraction under exit and voice. Section 8 derives the conservation principle for predation. Section 9 formalizes the legitimacy constraint. Section 10 treats justice as mechanism design. Section 11 provides evolutionary foundations. Section 12 assembles the dynamical system. Section 13 derives collapse and renewal. Section 14 addresses scope and objections. Section 15 concludes.

2 Primitives

2.1 Actors and Relations

Let $N = \{1, \ldots, n\}$ be a finite set of actors. A *relation* is an ordered pair $e = (i, j) \in E \subseteq N \times N$, $i \neq j$. Relations may be bilateral or, by extension, multilateral; the bilateral case is treated here for tractability. Each relation $e = (i, j)$ is characterized by:

- **Contributions** $c_i^e, c_j^e \in [0, \bar{c}]$, measured in units of reproductive capacity.
- **Output** $Q_e = A_e f(c_i^e, c_j^e)$, where $A_e > 0$ is a productivity parameter.
- **Shares** $s_i^e, s_j^e \geq 0$ with $s_i^e + s_j^e \leq 1$. The residual $1 - s_i^e - s_j^e$ is lost to friction or captured by a third party.
- **Exit cost** $k_i^e \geq 0$.
- **Voice** $v_i^e \geq 0$, an index of the actor's capacity to renegotiate terms.

2.2 Net Gain

Definition 2.1 (Net gain)**.** The *net gain* to actor i in relation e is

$$n_i^e \;=\; s_i^e Q_e - c_i^e. \tag{1}$$

The *surplus* of the relation is

$$S_e \;=\; Q_e - c_i^e - c_j^e. \tag{2}$$

2.3 Thresholds

Each actor possesses two thresholds.

Definition 2.2 (Participation threshold)**.** $\bar{a}_i \geq 0$ is the payoff actor i can obtain outside the relation. The relation is *individually rational* for i iff $n_i^e \geq \bar{a}_i$.

Definition 2.3 (Reproduction threshold)**.** $\rho_i > 0$ is the minimum net gain required for actor i to maintain its capacity. Formally, capacity K_i satisfies

$$\dot{K}_i \;=\; r_i K_i\Big(1 - \tfrac{K_i}{K_i^{\max}}\Big) \;+\; \sum_{e \ni i} n_i^e \;-\; \delta_i K_i \;-\; \phi_i C_i, \tag{3}$$

where r_i is the intrinsic growth rate, δ_i the depreciation rate, C_i conflict intensity, and $\phi_i > 0$ the damage coefficient. Actor i is *reproducing* iff $\dot{K}_i \geq 0$ along the balanced path.

Remark 2.4. The two thresholds are distinct. The participation threshold governs voluntary continuation; the reproduction threshold governs biological, material, and institutional survival. The gap between them is the space in which parasitism operates.

2.4 Assumptions

Assumption 2.5 (Diminishing returns)**.** For each relation, f is strictly increasing, strictly concave in each argument, and homogeneous of degree $\theta < 1$ in the long run, ensuring interior optima.

Assumption 2.6 (Bounded contributions)**.** $c_i^e \leq \bar{c} < \infty$.

Assumption 2.7 (Rationality)**.** Actors maximize discounted net gain with discount factor $\delta_i \in (0, 1)$ or discount rate $r_i > 0$.

3 Civilization as Process and Product

The theory developed in Section 2 treats civilization as an order of relations. This raises an immediate question: what of art, architecture, law, science, mathematics, engineering, philosophy, and literature? These are what the word "civilization" most often denotes in ordinary usage. The answer of this paper is that they are *products* of the relational order, not the order itself. Identifying civilization with its achievements is a category error—like identifying a factory with its output, or a city with its buildings.

3.1 Definition

Definition 3.1 (Civilizational achievement)**.** Let $\mathcal{D} = \{\text{art}, \text{architecture}, \text{law}, \text{science}, \text{mathematics}, \text{engineering}, \text{philosophy}, \text{literature}, \text{technology}, \dots\}$ be a set of domains. A *civilizational achievement* is a vector $A = (A_d)_{d \in \mathcal{D}}$, where $A_d \geq 0$ measures the accumulated stock of durable products in domain d.

Definition 3.2 (Achievement production function)**.** Achievements are produced by the relational order. Write

$$A_d \;=\; G_d\big(S,\; \delta,\; \theta,\; H\big), \qquad \frac{\partial G_d}{\partial S}, \frac{\partial G_d}{\partial \delta}, \frac{\partial G_d}{\partial \theta}, \frac{\partial G_d}{\partial H} \;\geq\; 0, \tag{4}$$

where

- $S = \sum_e S_e$ is aggregate surplus from eq. (2);
- δ is the patience parameter of Section 6;
- θ is organizational capacity (coordination, administration, institutional continuity);
- H is the accumulated stock of human capital and knowledge.

Remark 3.3. The production function is *relational*, not technological. The inputs S, δ, θ, H are themselves consequences of the ledger: surplus depends on contributions and shares; patience depends on the expected duration of relations; organizational capacity depends on institutions; human capital depends on who is permitted to learn. Achievements are therefore downstream of the relational order, not upstream of it.

3.2 Propositions

Proposition 3.4 (Surplus is necessary)**.** *If $S \leq 0$, then $A_d = 0$ for all d that require accumulation.*

Proof. By eq. (4), G_d is nondecreasing in S and vanishes when the economy is at subsistence. A subsistence economy has no surplus to allocate to non-reproductive activity. □

Proposition 3.5 (Achievement is independent of justice)**.** *There exist parasitic and predatory equilibria with arbitrarily large A.*

Proof. Choose an equilibrium with high S concentrated in the hands of a principal, high δ, and high θ. By Proposition 7.2 the principal can extract up to τ^{exit}; by Proposition 8.1 a patient predator conserves the host. Either way, the principal commands surplus and patience, and can commission achievements. The classification of Section 4 places such an equilibrium in the parasitic or predatory category. Hence A can be large while $n_j^e < \bar{a}_j$ for the host. □

Corollary 3.6 (Achievement is not evidence of symbiosis)**.** *The presence of monuments, codes, temples, and treatises does not license the inference that the underlying order was symbiotic.*

Illustration 3.7. Egyptian pyramids, Roman roads, Athenian temples, and Mesoamerican pyramids were all produced under parasitic or predatory relations. Athenian philosophy coexisted with chattel slavery; Roman law coexisted with provincial extraction; Egyptian geometry coexisted with corvée. The achievements are real; the inference from achievement to justice is not.

Proposition 3.8 (Achievement is diagnostic of the ledger)**.** *The composition of A—which domains are funded, by whom, and for whom—is a signature of the relational order that produced it.*

Proof. Each A_d requires inputs (S, δ, θ, H) that are allocated through the ledger. The choice of d reveals the preferences of those who control the surplus; the scale of A_d reveals the size of the surplus and the patience of its controllers; the durability of A_d reveals the time horizon δ. Monuments are therefore fossilized ledgers. □

Remark 3.9. This proposition converts achievement from a criterion of civilization into a *diagnostic instrument* for studying it. The question is not "how advanced is this civilization?" but "who commanded the labor, who bore the cost, who consumed the product, and what does the pattern of achievement reveal about the shares s_i^e?"

Proposition 3.10 (Common cause)**.** *Patience δ raises both the durability of achievement and the sustainability of extraction. High-δ orders are simultaneously the most creative and the most exploitative.*

Proof. By Proposition 6.5, δ raises the sustainability of cooperation. By Proposition 8.1, low $r = 1 - \delta$ raises the conservation of the host. By eq. (4), δ raises A. The same parameter that permits long-horizon investment permits long-horizon extraction. □

Illustration 3.11. The pharaoh's corvée and the pharaoh's pyramids share the same δ. The Roman *pax* and the Roman aqueduct share the same δ. The Mandate of Heaven and the Grand Canal share the same δ. Achievements and extraction are not opposed; they are joint products of the same temporal structure.

Proposition 3.12 (Lagged collapse)**.** *If the relational order collapses, A collapses with a lag. The lag is the cultural residue: achievements outlive the relations that produced them.*

Proof. By Theorem 13.1, S, δ, θ, H decline after collapse. By eq. (4), A declines in these arguments. Because A is a stock and not a flow, it decays at the depreciation rate rather than instantaneously. □

Corollary 3.13 (Ruins as evidence)**.** *Ruins are evidence of a past relational order, not of a present one. They are the residue of a ledger that has already been settled—or defaulted.*

3.3 The Category Error

Remark 3.14 (Against the monumental definition). The historiographic habit of identifying civilization with its achievements is not merely imprecise; it is systematically misleading. It selects on the dependent variable. It notices pyramids but not the peasants who built them; it celebrates codes but not the parties they bound; it catalogs temples but not the tithes that funded them. The monumental definition answers "what did they build?" The relational definition answers "who gave, who gained, who lost, and how was it justified?" Only the latter explains why the achievements appeared, why they took the form they did, and why they collapsed when they did.

Remark 3.15 (Against the progress narrative). If achievements are products of the ledger, then the history of civilization is not a history of increasing enlightenment. It is a history of shifting terms of reciprocity. The same century may produce both a philosophical masterpiece and an imperial atrocity, because both flow from the same surplus, patience, and organizational capacity. Progress in A does not entail progress in justice.

3.4 Implications

The distinction between process and product yields three methodological commitments.

1. **Measure the ledger first.** Before cataloging achievements, identify the relations, the shares, the thresholds, and the justifications.
2. **Treat achievements as evidence, not criteria.** A code, a temple, or a treatise is data about the order that produced it, not a measure of its justice.
3. **Look for the excluded.** The most revealing achievements are often those that required the most invisible labor: monuments, walls, canals, and mines.

The remainder of the paper develops the formal apparatus for this program.

4 The Classification of Relations

4.1 Definitions

Definition 4.1 (Symbiosis)**.** Relation $e = (i, j)$ is *symbiotic* iff

$$n_i^e \geq \bar{a}_i \quad \text{and} \quad n_j^e \geq \bar{a}_j.$$

Both parties find continued participation preferable to exit.

Definition 4.2 (Parasitism)**.** Relation $e = (i, j)$ is *parasitic* iff, for some j,

$$\rho_j \;\le\; n_j^e \;<\; \bar{a}_j,$$

and $n_i^e > \bar{a}_i$. Party j is a *host*: it is exploited below its participation threshold but above its reproduction threshold.

Definition 4.3 (Predation)**.** Relation $e = (i, j)$ is *predatory* iff, for some j,

$$n_j^e < \rho_j.$$

Party j's capacity declines; the host is being consumed.

Definition 4.4 (Mixed order)**.** A civilization $\Gamma = (N, E)$ is *mixed* if E contains relations of more than one type.

4.2 The Classification Proposition

Proposition 4.5. *The three categories in Definitions 4.1 to 4.3 partition the space of bilateral relations by the ordered pair (n_i^e, n_j^e) and the thresholds $(\bar{a}_i, \bar{a}_j, \rho_i, \rho_j)$.*

Proof. The thresholds define three intervals for each actor on the real line: $(-\infty, \rho_i)$, $[\rho_i, \bar{a}_i)$, and $[\bar{a}_i, \infty)$. The Cartesian product of two such partitions yields nine cells; collapsing by the identity of the exploited party yields the three categories together with their symmetric and degenerate cases. □

Remark 4.6. The classification is not moral. A parasitic relation may be legal, religiously sanctioned, and stable for centuries. A predatory relation may be celebrated as conquest. The categories describe the sign and sustainability of net transfers, not their legitimacy.

Illustration 4.7. Mesopotamian corvée under the temple economy was parasitic on peasants while reproducing the irrigation infrastructure; Athenian chattel slavery was predatory on the enslaved; the Delian League was parasitic on allied cities; the *andurarum* was a mechanism that converted predatory debt relations back into parasitic and eventually symbiotic ones.

5 The Viability Set

5.1 The Cobb–Douglas Case

Let $f(c_i, c_j) = c_i^{\alpha} c_j^{\beta}$ with $\alpha, \beta > 0$ and $\alpha + \beta < 1$. Define $D = 1 - \alpha - \beta > 0$.

Lemma 5.1 (Interior optimum)**.** *The unique interior Nash equilibrium of the one-shot relation game is*

$$c_i^* = (s_i \alpha A)^{\frac{1-\beta}{D}} (s_j \beta A)^{\frac{\beta}{D}}, \qquad c_j^* = (s_j \beta A)^{\frac{1-\alpha}{D}} (s_i \alpha A)^{\frac{\alpha}{D}}, \tag{5}$$

with equilibrium payoffs

$$u_i^* = s_i A (1 - \alpha)\, (s_i \alpha A)^{\alpha/D} (s_j \beta A)^{\beta/D}. \tag{6}$$

Proof. First-order conditions yield $c_i^{1-\alpha} = s_i \alpha A c_j^{\beta}$ and $c_j^{1-\beta} = s_j \beta A c_i^{\alpha}$. Substituting and solving the resulting linear system in logarithms gives eq. (5). Substitution into $u_i = s_i Q - c_i$ and factoring yields eq. (6). □

5.2 The Viability Set

Definition 5.2 (Viability set)**.** The *viability set* of a relation is

$$\mathcal{V} = \big\{(s_i, s_j) \in \Delta^2 : \; u_i^* \geq \bar{a}_i, \; u_j^* \geq \bar{a}_j\big\}, \tag{7}$$

where $\Delta^2 = \{(s_i, s_j) : s_i, s_j \geq 0, \; s_i + s_j \leq 1\}$.

Proposition 5.3 (Monotonicity of viability)**.** *For fixed s_j, u_i^* is strictly increasing in s_i. Hence $\mathcal{V}$ is an interval in each share and is nonempty whenever $\bar{a}_i, \bar{a}_j$ are sufficiently small.*

Proof. From eq. (6), $\log u_i^* = \log s_i + \log A(1-\alpha) + \frac{\alpha}{D}\log(s_i \alpha A) + \frac{\beta}{D}\log(s_j \beta A)$. Differentiating with respect to s_i gives $\frac{1}{s_i} + \frac{\alpha}{D s_i} > 0$. □

Corollary 5.4. *There exist thresholds $\underline{s}_i(\bar{a}_i)$ and $\underline{s}_j(\bar{a}_j)$ such that $(s_i, s_j) \in \mathcal{V}$ iff $s_i \geq \underline{s}_i$ and $s_j \geq \underline{s}_j$.*

5.3 The Emergence of Parasitism

Proposition 5.5 (Necessity of constraint)**.** *If both parties choose contributions freely and shares lie in $\mathcal{V}$, the relation is symbiotic. Parasitism cannot arise from voluntary contribution choice alone.*

Proof. From eq. (6), $u_i^* > 0$ for all $s_i, s_j > 0$. Free choice therefore always yields nonnegative net gain. Parasitism requires either that shares fall outside $\mathcal{V}$ or that contributions are constrained. □

Remark 5.6. This is a substantive result. Parasitism is not a failure of rationality; it is a consequence of *constraint*. The constraints are historically specific: contribution floors (corvée, slavery, debt peonage), exit costs (serfdom, closed borders, indenture), and suppressed voice (disenfranchisement, censorship, martial law).

Definition 5.7 (Contribution floor)**.** A relation imposes a floor $c_j^e \geq \underline{c}_j > 0$ on the host.

Proposition 5.8 (Parasitic equilibrium)**.** *With a contribution floor $\underline{c}_j$, the constrained optimum satisfies*

$$u_j^{con} = s_j Q(\underline{c}_j) - \underline{c}_j. \tag{8}$$

The relation is parasitic iff $u_j^{con} \in [\rho_j, \bar{a}_j)$ and predatory iff $u_j^{con} < \rho_j$.

6 Repetition and the Sustainability of Cooperation

6.1 The Repeated Relation Game

Let G be the one-shot relation game of Section 5, repeated infinitely with common discount factor $\delta \in (0, 1)$.

Definition 6.1. The *cooperative payoff* is $u_i^C = u_i^*$ from eq. (6). The *defection payoff* is $u_i^D = \max_{c_i} s_i Q(c_i, c_j^*) - c_i$. The *punishment payoff* is u_i^P, the stage payoff under reversion to the one-shot equilibrium of the punishment phase.

Proposition 6.2 (Sustainability of symbiosis)**.** *The symbiotic profile is a subgame-perfect equilibrium of the repeated game iff*

$$\delta \; \geq \; \delta_i^* \; = \; \frac{u_i^D - u_i^C}{u_i^D - u_i^P} \quad \textit{for all } i. \tag{9}$$

Proof. Standard grim-trigger argument. Cooperation is optimal iff the one-period gain from defection, $u_i^D - u_i^C$, does not exceed the discounted present value of the punishment, $\frac{\delta}{1-\delta}(u_i^C - u_i^P)$. Rearranging yields eq. (9). □

6.2 The Patience Threshold

Definition 6.3. $\delta^* = \max_i \delta_i^*$ is the *patience threshold of civilization.*

Remark 6.4. δ^* is not a psychological parameter. It is an institutional one. Long time horizons arise from repeated interaction, secure property, dynastic succession, corporate continuity, legal personality, and the expectation of future dealings. Institutions raise δ; anarchy lowers it.

Proposition 6.5 (Institutional comparative statics)**.** *δ^* is decreasing in the expected duration of the relation, in the observability of contributions, and in the severity of punishment; it is increasing in the value of the outside option and in the anonymity of interaction.*

Proof. Each claim follows by differentiating eq. (9) with respect to the relevant parameter under the standard comparative statics of the repeated game. □

Illustration 6.6. The *andurarum* raised δ by resetting obligations and restoring the expectation of future cooperation. Ma'at raised δ by sacralizing long horizons. Roman law raised δ by making contracts enforceable across generations. The Mandate of Heaven lowered δ as the dynasty aged: the expected remaining duration of the relation fell.

7 Extraction, Exit, and Voice

7.1 The Extraction Game

A principal P sets an extraction rate $\tau \in [0, 1]$. An agent A chooses effort $e \in [0, 1]$. Output $Q = e$. Payoffs:

$$u_P = \tau e, \qquad u_A = (1 - \tau)e - \tfrac{1}{2}e^2. \tag{10}$$

Lemma 7.1. *The agent's best response is $e^* = 1 - \tau$, yielding*

$$u_A(\tau) = \tfrac{1}{2}(1 - \tau)^2, \qquad u_P(\tau) = \tau(1 - \tau). \tag{11}$$

7.2 The Unconstrained Optimum

Proposition 7.2 (Revenue-maximizing extraction)**.** *Absent exit, the principal chooses $\tau^{mon} = 1/2$.*

Proof. Maximize $\tau(1 - \tau)$. □

7.3 The Exit Constraint

Proposition 7.3 (Exit ceiling)**.** *If the agent can exit at cost k, the principal's extraction is bounded by*

$$\tau^{exit} = 1 - \sqrt{2(\bar{a}_A + k)}, \tag{12}$$

and the equilibrium rate is $\tau^* = \min\{\tau^{mon}, \tau^{exit}\}$.

Proof. The agent exits iff $u_A(\tau) < \bar{a}_A + k$. Substituting eq. (11) and solving for τ yields eq. (12). □

Corollary 7.4. *High exit costs raise the extraction ceiling. Serfdom, indenture, closed borders, and debt bondage are technologies for raising k.*

7.4 The Voice Constraint

Definition 7.5. Voice $v_A \geq 0$ is the agent's capacity to impose costs on the principal conditional on disagreement.

Proposition 7.6 (Nash bargaining with voice)**.** *Under generalized Nash bargaining with disagreement payoffs (d_P, d_A) and bargaining power $\theta = v_A/(v_A + v_P)$, the equilibrium extraction rate is*

$$\tau^{NB} = \arg\max_{\tau} \left[\tau(1-\tau) - d_P\right]^{\theta} \left[\tfrac{1}{2}(1-\tau)^2 - d_A\right]^{1-\theta}. \tag{13}$$

Moreover $\partial\tau^{NB}/\partial\theta > 0$: extraction rises with the principal's bargaining power.

Remark 7.7. Voice is not merely moral suasion. It is the capacity to withhold cooperation, to strike, to litigate, to vote, to riot, to secede. It is an economic magnitude.

Illustration 7.8. Athenian liturgies were voluntary τ by elites, sustained by honor and civic reciprocity. Roman provincial tribute was τ at the exit ceiling for provinces without voice. The Chinese examination system was an institutional mechanism for raising the voice of the gentry, thereby lowering extraction below the rebellion threshold.

8 Predation and the Conservation of the Host

8.1 The Optimal Extraction Problem

Let K denote host capacity, growing logistically:

$$\dot{K} = K(1-K) - \tau K, \tag{14}$$

with extraction τ and discount rate $r > 0$. The predator maximizes

$$\max_{\{\tau(t)\}} \int_0^{\infty} e^{-rt}\, \tau K \, dt. \tag{15}$$

Proposition 8.1 (Predation principle)**.** *The optimal steady-state host stock is*

$$K^*(r) = \frac{1-r}{2}, \tag{16}$$

which is strictly decreasing in r. A predator with $r \geq 1$ depletes the host entirely.

Proof. The current-value Hamiltonian is $H = \tau K + \mu[K(1-K) - \tau K]$. The singular control satisfies $\partial H/\partial\tau = K(1-\mu) = 0$, hence $\mu = 1$. The costate equation gives $\dot{\mu} = r\mu - \partial H/\partial K$; on the singular arc $\dot{\mu} = 0$, yielding $r = 1 - 2K$. Solving gives eq. (16). □

8.2 Interpretation

Corollary 8.2 (Conservation)**.** *The more patient the predator, the more it conserves the host. Empire—a long-horizon predator—builds infrastructure, administers justice, and invests in the productive capacity of the periphery. Raiding—a short-horizon predator—loots and departs. The difference between Rome and a Viking raid is not morality but r.*

Corollary 8.3 (Predation threshold)**.** *Predation becomes parasitic rather than destructive precisely when the predator's discount rate falls below the host's intrinsic growth rate.*

9 Legitimacy and the Cost of Compliance

9.1 The Compliance Technology

Definition 9.1. Let $G = \sum_i \max(0, \bar{a}_i - n_i)$ denote the aggregate *injustice gap*. Compliance is maintained at cost

$$\Phi = \Phi(G, L), \qquad \Phi_G > 0,\ \Phi_L < 0,\ \Phi_{GG} > 0, \tag{17}$$

where L is legitimacy.

Definition 9.2. An order is *feasible* iff

$$\sum_e \tau_e Q_e \ \geq\ \Phi(G, L) + M, \tag{18}$$

where M is the fixed cost of administration.

9.2 The Legitimacy Constraint

Proposition 9.3 (Maximum sustainable injustice)**.** *Under eqs.* (17) *and* (18)*, there exists a finite $\bar{G}$ such that no extraction level sustains the order whenever $G > \bar{G}$.*

Proof. Revenue is bounded above by $\bar{\tau}\bar{Q}$. Feasibility requires $\Phi(G, L) \leq \bar{\tau}\bar{Q} - M$. Since Φ is strictly increasing and convex in G, there exists $\bar{G}$ with $\Phi(\bar{G}, L) = \bar{\tau}\bar{Q} - M$. □

Corollary 9.4 (Repression is expensive)**.** *A parasitic equilibrium sustained by coercion alone requires $\Phi(G, 0)$, which is strictly greater than $\Phi(G, L)$ for $L > 0$. Legitimacy is a cost-reducing technology.*

Corollary 9.5 (Legitimacy decay)**.** *If $\dot{L} = -\lambda G + \mu I$, where I is ideological investment, then sustained injustice erodes the very legitimacy that cheapens compliance. Parasitic equilibria are therefore dynamically fragile even when statically stable.*

Illustration 9.6. Ma'at functioned as an ideological input I that raised L and thereby lowered the cost of corvée compliance. The Mandate of Heaven performed the analogous function in China, and its withdrawal was precisely a collapse of L below the threshold at which eq. (18) could hold.

9.3 Religion as a Legitimacy Technology

The framework appears to leave religion unexplained. It does not. It assigns religion to the legitimacy technology of this section, and it identifies the structural reason why religion both *solidifies* an established order and can be used to *overturn* one.

Definition 9.7 (The divine counterparty)**.** A *religious relation* is a relation whose counterparty D is unobservable: no capacity K_D, reproduction threshold ρ_D, participation threshold $\bar{a}_D$, exit cost k_D, or voice v_D can be measured, and the output promised to the human party is not verifiable. In the tuple of Section 2, the divine side is formally blank.

Proposition 9.8 (Indeterminacy on the divine side)**.** *Because the net transfer n^D of a religious relation cannot be measured, the classification of Definition 4.1, Definition 4.2, and Definition 4.3 is undefined for the divine side. The theory neither affirms nor denies the relation; it registers instead the relation's* earthly *counterpart—the transfer between worshipper and institution—which is an ordinary relation and is classified accordingly.*

Proof. Classification requires a measurable net transfer for each party. The divine transfer is unobservable by Definition 9.7, so the predicate is undefined on that side; the human-to-institution transfer remains well defined. □

Proposition 9.9 (The inexhaustible host)**.** *If D is modeled with $K_D = \infty$ and $\dot{K}_D = 0$, the conservation corollary of Proposition 8.1 is vacuous: the host cannot be depleted, so the relation can never be classified parasitic or predatory, however much the human party contributes. An inexhaustible and non-substitutable counterparty is the unique way to escape the conservation constraint, and it renders the relation permanently symbiotic in appearance.*

Proof. The predation condition requires $\dot{K}_D < 0$ at positive extraction. $K_D = \infty$ denies it, so the condition never binds. □

Corollary 9.10 (Solidification)**.** *A religious relation contributes ideological input $I > 0$ to eq.* (24)*, raising L and lowering the compliance cost $\Phi(G, L)$ of eq.* (17)*. It is a cost-reducing technology for every extraction it legitimizes. This is why religion solidifies an order: it does not add force, it subtracts the price of force.*

Proposition 9.11 (Reformation)**.** *Let an incumbent order be feasible at L, and let L^* be the threshold below which eq.* (18) *fails. A rival legitimacy technology that lowers the incumbent's I—by reinterpreting or discrediting D—lowers L and can drive $L < L^*$ without any change in the balance of force. A new religion is thus a vehicle for replacing an old order: it reallocates the legitimacy stock rather than conquering it.*

Proof. From eq. (24), L is increasing in I; lowering I lowers L. Proposition 9.3 gives a finite $\bar{G}$ at each L, so a sufficient fall in L makes $\Phi(G, L)$ exceed any feasible revenue. □

Remark 9.12 (Deception, or the signal monopoly). Because the counterparty's output is unverifiable, the human intermediary holds a *signal monopoly*: it alone interprets what D demands and grants. The religious relation is therefore a principal–agent problem with both hidden action and hidden information, in which the intermediary can invoke D to justify any transfer τ. This is the formal content of the observation that power manipulates religion: the indeterminacy that makes religion a cheap legitimacy technology is the same indeterminacy that makes it manipulable. The theory does not claim that D is false. It claims that the ledger cannot audit the divine side, so the audit must fall on the earthly institution that speaks for it.

9.4 The Taxonomy of Legitimacy Technologies

Religion is one legitimacy technology among several. It is worth naming the others, because they are substitutes and an order selects among them.

Definition 9.13 (Legitimacy technology)**.** A *legitimacy technology* is any intervention that raises L or lowers $\Phi(G, L)$ per unit of cost, other than direct repression R. Equivalently, it is an input to the dynamics of L or to the arguments of Φ other than R itself.

Table 1: A taxonomy of legitimacy technologies.

Type	Mechanism	Acts on	Examples
Belief / ideological	supplies I	L	religion, nationalism, political ideology, meritocracy, "the market," progress, tradition, charisma
Procedural / voice	raises F, v; lowers k	G, D	due process, courts, elections, transparency, rights, federalism
Performance / output	delivers material goods	G	growth, welfare residue, security, public goods, crisis competence
Coercion-substitute	raises R; fragments X	Φ	surveillance, predictive compliance, atomization, credit scoring, platform curation
Temporal / institutional	raises δ; lowers r	extraction, G	dynastic continuity, legal personality, property rights, credible commitment, calendars
Ceremonial / spectacular	recruits products as signals	L	monuments, festivals, sport, mass culture

The classical sociology anticipates the belief row: Weber's traditional, charismatic, and legal-rational authority; Scharpf's input and output legitimacy. The coercion-substitute row is the distinctively modern move—it substitutes for belief rather than supplying it.

Definition 9.14 (Auditability)**.** A legitimacy technology is *auditable* to the degree that the claims it uses to justify a transfer τ can be verified by the parties who bear τ.

Proposition 9.15 (Verifiability disciplines)**.** *The flexibility of a legitimacy technology for justifying arbitrary transfers is decreasing in its auditability. An unverifiable technology grants its interpreter a signal monopoly; a verifiable technology constrains the interpreter by checkable fact.*

Proof. Justifying τ requires a claim whose truth the bearer cannot contest. If the claim is verifiable, a false claim is detectable and contestable, which raises the cost Φ of imposing τ; if the claim is unverifiable, no such detection is possible, and the interpreter retains full latitude. The continuum of judgment is therefore indexed by auditability. □

Corollary 9.16 (The auditability ordering)**.** *Religion, whose counterparty is unobservable (Definition 9.7), is maximally flexible and hence maximally manipulable. Science, whose claims are costly to fake and cheap to check, is minimally so. Law, markets, and elections lie between: their procedures are auditable, but their "justice," "value," and "mandate" claims are not, and manipulation concentrates at that seam.*

This generalizes Remark 9.12. The same move—invoke an unverifiable authority to justify a transfer—appears across rows: divine will, national destiny, "the science is settled," "the market has spoken," "the algorithm decided." Because the technologies are substitutes, an order selects the cheapest available mix, and the mix shifts over time—from religion to nationalism to output legitimacy to architecture.

10 Justice as Mechanism Design

10.1 The Planner's Problem

A planner with welfare weights $\lambda_i > 0$ solves

$$\max_{\{s_i, T_{ij}\}} \sum_i \lambda_i u_i \quad \text{s.t.} \quad \text{(IC)} \;\; u_i \geq u_i^D, \qquad \text{(IR)} \;\; u_i \geq \bar{a}_i, \qquad \sum_i T_{ij} = 0. \tag{19}$$

Proposition 10.1 (Rebalancing rule)**.** *At an interior optimum, the planner equates weighted marginal utilities of net gain across actors:*

$$\lambda_i \frac{\partial u_i}{\partial s_i} = \lambda_j \frac{\partial u_j}{\partial s_j}. \tag{20}$$

When $\lambda_i = \lambda_j$, the optimum requires $s_i = s_i^{fair}$, where

$$s_i^{fair} = \arg\max_{s_i} [u_i(s_i) + u_j(1 - s_i)]. \tag{21}$$

Proposition 10.2 (Justice as gap reduction)**.** *Define the injustice gap $G_i = \max(0, u_i^{fair} - u_i)$. Justice mechanisms reduce G_i through four instruments:*

1. ***Transfers:*** *$T_{ij} > 0$ from gainers to losers.*
2. ***Rights:*** *constraints on τ.*
3. ***Procedures:*** *increases in v_i and reductions in k_i.*
4. ***Repair:*** *restoration of K_i after predation.*

Proof. Each instrument enters the constraint set of eq. (19) and relaxes it. □

10.2 The Adjustment Correspondence

Definition 10.3. A *justice regime* is a map $J : \mathcal{S} \to \mathcal{S}$ from states of the ledger to states of the ledger.

Proposition 10.4 (Effectiveness)**.** *A justice regime is effective iff J reduces G faster than the disequilibrium forces of Section 12 increase it.*

Illustration 10.5. *Andurarum* was a discrete reset. Solon's *seisachtheia* was a one-time rebalancing. The Chinese land reforms were periodic. The Roman extension of citizenship was incremental. Effective regimes may be discrete or continuous; what matters is that J outruns the drift.

11 Evolutionary Foundations

11.1 The Stage Game

Consider a population playing a symmetric relation game with strategies C (cooperate), E (exploit), and M (punish exploiters). Let $b > c > 0$ be benefit and cost of cooperation, $w > 0$ the cost of mutual conflict, $p > 0$ the cost of punishing, and $\pi > 0$ the fine imposed on an exploiter.

Table 2: Payoff matrix of the evolutionary relation game.

	C	E	M
C	$b - c$	$-c$	$b - c$
E	b	$-w$	$b - \pi$
M	$b - c$	$-c - p$	$b - c$

11.2 Evolutionary Stability

Proposition 11.1 (Institutional stabilization of cooperation)**.** *C is an evolutionarily stable strategy iff*

$$\pi > c + t \qquad \textit{and} \qquad w > c + p, \tag{22}$$

where t is the tax financing punishment.

Proof. C is an ESS iff $u(C,C) > u(E,C)$ and $u(C,E) > u(E,E)$. The first inequality gives $b - c > b - \pi$, i.e., $\pi > c$. With institutional financing at rate t, it becomes $\pi > c + t$. The second gives $-c - p > -w$, i.e., $w > c + p$. □

Remark 11.2. Cooperation is evolutionarily stable only when punishment is strong enough to deter exploitation and conflict is costly enough that a lone cooperator fares better than a pair of exploiters. Civilization is the institutional manufacture of these two conditions.

Corollary 11.3 (Parasitism as polymorphism)**.** *When $\pi < c + t$ but w is large, C and E coexist at a mixed equilibrium. This is the formal counterpart of the stable agrarian empire: a large class of cooperators, a small class of extractors, and institutions that prevent the extractors from consuming the cooperators.*

Corollary 11.4 (Predation is transient)**.** *Because predation destroys the host and hence the payoff b, it cannot be an ESS in any environment where hosts are finite.*

12 The Dynamical System

12.1 State Variables

For each actor i, the state is (K_i, L_i, C_i, s_i). The dynamics are

$$\dot{K}_i = r_i K_i\Big(1 - \tfrac{K_i}{K_i^{\max}}\Big) + \sum_{e \ni i} n_i^e - \delta_i K_i - \phi_i C_i, \tag{23}$$

$$\dot{L}_i = \alpha(F_i - F^*) - \beta R_i + \gamma I_i, \tag{24}$$

$$\dot{C}_i = \sigma\big(D_i + P_i - \lambda L_i - \mu X_i\big), \tag{25}$$

$$\dot{s}_i = \kappa\big(s_i^{\text{fair}} - s_i\big), \tag{26}$$

where F_i is perceived fairness, R_i repression, I_i ideology, D_i relative deprivation, P_i power asymmetry, X_i exit/voice options, and σ a sigmoid.

12.2 Equilibrium Classification

Definition 12.1. A state is a **symbiotic equilibrium** iff $n_i^e \geq \bar{a}_i$ for all i and e, $C = 0$, and $\dot{K} \geq 0$. It is a **parasitic equilibrium** iff $n_j^e \in [\rho_j, \bar{a}_j)$ for some j, $C = 0$, and $\dot{K} \geq 0$. It is **predatory** iff $n_j^e < \rho_j$ for some j.

Proposition 12.2 (Stability)**.** *A symbiotic equilibrium is locally asymptotically stable iff the Jacobian of eqs.* (23) *to* (26) *has eigenvalues with negative real part. A parasitic equilibrium is stable iff the repression and ideological terms in eq.* (24) *dominate the fairness gap. A predatory equilibrium is generically unstable.*

Proof. The first claim is the definition of local stability. The second follows from linearizing eqs. (24) and (25): stability requires $\lambda L_i + \mu X_i > D_i + P_i$. The third follows from the fact that $\dot{K}_j < 0$ drives $K_j \to 0$, reducing future Q and hence u_i, contradicting stationarity. □

12.3 Phase Transitions

Proposition 12.3 (Transition thresholds)**.** *The system exhibits the following transitions:*

1. ***Symbiosis → parasitism:*** *s_i falls below $\underline{s}_i$.*
2. ***Parasitism → predation:*** *K_i falls below K_i^{crit}.*
3. ***Predation → collapse:*** *$K_i \to 0$, $L_i \to 0$, $C_i \to \infty$.*
4. ***Collapse → renewal:*** *institutional reset restores s_i, L_i, K_i.*

Proof. Each transition corresponds to the crossing of a threshold identified in Propositions 5.8, 8.1 and 9.3. □

13 Collapse and Renewal

13.1 The Collapse Theorem

Theorem 13.1 (Collapse)**.** *Suppose (i) G increases monotonically with extraction τ; (ii) $\dot{L} = -\lambda G$; (iii) C increases in G and decreases in L; (iv) $\dot{K}$ decreases in C; and (v) no justice mechanism operates. Then there exists a finite time T at which the order collapses.*

Proof. By (i)–(ii), sustained extraction drives L monotonically to zero. By (iii), C then increases without bound. By (iv), K declines monotonically. Since $K \geq 0$, either $K \to 0$ or C diverges; in either case the order cannot be sustained. □

Corollary 13.2 (Necessity of justice)**.** *Any parasitic equilibrium is ultimately unsustainable in the absence of a justice mechanism.*

13.2 The Dynastic Cycle

Proposition 13.3 (Oscillation)**.** *Let the justice mechanism respond to the injustice gap with lag:*

$$\dot{G} = a - bJ, \qquad \dot{J} = c(G - G_0), \tag{27}$$

with $a, b, c > 0$. The system has a center at $(G_0, a/b)$ with eigenvalues $\pm i\sqrt{bc}$, and therefore oscillates.

Proof. Linearize about the fixed point. The Jacobian is $\begin{pmatrix} 0 & -b \\ c & 0 \end{pmatrix}$ with characteristic polynomial $\lambda^2 + bc = 0$. □

Remark 13.4. With nonlinearities and saturation, the center becomes a limit cycle—the formal counterpart of the dynastic cycle, the rise and fall of empires, and the recurrence of debt crises and jubilees.

Illustration 13.5. Mesopotamian debt cancellation, Chinese dynastic renewal, and Roman citizenship extension are instances of the reset mechanism J. Their periodic recurrence is not coincidence; it is the signature of a relaxation oscillator.

14 Scope and Objections

14.1 The Marxist Objection

Marxists will object that "win–win" rhetoric masks exploitation. The theory agrees. The classification of Section 4 makes exploitation a first-class category, and the legitimacy constraint of Section 9 shows why exploitation is unstable. The theory does not claim that civilizations are just; it claims that they must *manage* injustice.

14.2 The Realist Objection

Realists will object that power, not reciprocity, governs international relations. The theory agrees that power shapes shares. But Proposition 7.3 shows that power is bounded by exit, and Proposition 9.3 shows that it is bounded by legitimacy. Power determines where in the feasible set the equilibrium lies; it does not eliminate the feasible set.

14.3 The Culturalist Objection

Culturalists will object that religion, honor, and identity cannot be reduced to gain and loss. The theory treats culture as a justification regime, formalized in the legitimacy variable L and the ideological input I. Culture enters the model, but as an argument, not as an alternative. The theory also holds that cultural achievements themselves are products of the ledger, as Section 3 argues, not autonomous causes of it.

14.4 The Functionalist Objection

Functionalists will object that stability proves fitness. The theory rejects the inference. A parasitic equilibrium is stable until it is not; Proposition 12.2 and Theorem 13.1 show why.

14.5 The Definitional Objection

"Civilization" carries colonial baggage. The theory uses it analytically, to denote a durable institutionalized order of relations. It makes no claim about hierarchy or superiority. Section 3 reinforces this point by severing the link between civilization and its monumental products.

14.6 The Limits of Metaphor

Biological metaphors—symbiosis, parasitism, predation—describe flows of costs and benefits, not essences. Human relations are reflexive: actors can renegotiate, revolt, and transform the terms of their association. The model treats this reflexivity as the justice mechanism J.

14.7 Empirical Scope

The theory is a framework, not a predictive model. Its value lies in specifying what must be measured: contributions, benefits, shares, thresholds, exit costs, voice, legitimacy, conflict, capacity. It generates comparative hypotheses—that orders with strong justice mechanisms are more durable, that exit costs raise extraction, that patient predators conserve, that legitimacy reduces the cost of compliance, that achievements are joint products of the same parameters that sustain extraction—which are testable against the historical record.

15 Conclusion

Civilization is a ledger of human relations. Every durable order must solve a reciprocity problem: it must align contributions with benefits closely enough that participants reproduce their capacity to participate. Relations are symbiotic when both parties reproduce, parasitic when one extracts from a surviving host, and predatory when extraction destroys the host. Stability does not prove justice; it may only prove that the losers cannot leave and cannot speak. The theory yields a small set of general results. Symbiosis is sustainable when patience is high, monitoring is possible, and punishment is credible. Parasitism is stable when exit is costly and voice is suppressed, but it is bounded by a legitimacy constraint and vulnerable to shocks. Predation is self-limiting, because consuming the host destroys the basis of future extraction. Justice is mechanism design—the institutional rebalancing of the ledger—and its effectiveness

determines the durability of the order. Cooperation is not evolutionarily stable in the state of nature; it must be institutionally manufactured. War and conflict are bargaining failures: the signal that rebalancing has failed. Collapse and renewal are phase transitions in a dynamical system: the signature of an oscillator that has not yet found its equilibrium. The achievements of a civilization—its art, architecture, law, science, mathematics, and engineering—are products of its relational order, not the order itself. They are produced by surplus, patience, organizational capacity, and human capital, all of which are themselves consequences of the ledger. Their presence is therefore no evidence of justice, and their absence no evidence of injustice. They are diagnostic, not constitutive. The historian who wishes to understand a civilization must ask not only what it built, but who was commanded to build it, who bore the cost, and who consumed the product. The theory does not replace historical judgment. It disciplines it. It tells us what to measure, what to ask, and what to hope for: a symbiotic equilibrium, sustained by institutions of justice, in which cooperation is not merely rational but reciprocal.

A Summary of Formal Results

Table 3: Summary of formal results.

Result	**Statement**	**Interpretation**
Surplus is necessary	$S \leq 0 \Rightarrow A_d = 0$	No surplus, no achievement
Achievement is independent of justice	Parasitic/predatory equilibria with large A exist	Monuments are not evidence of justice
Achievement is diagnostic	A reflects (S, δ, θ, H)	Monuments are fossilized ledgers
Common cause	δ raises both A and sustainable extraction	Creativity and exploitation are joint products
Lagged collapse	A decays at depreciation rate after collapse	Ruins are residues of settled ledgers
Interior optimum	Equations (5)–(6)	Interior equilibrium of the relation game
Monotonicity of viability	u_i^* increasing in s_i	Viability is an interval
Necessity of constraint	Free choice cannot produce parasitism	Constraints are the source of extraction
Parasitic equilibrium	$u_j^{\text{con}} = s_j Q(\underline{c}_j) - \underline{c}_j$	Contribution floors generate parasitism
Sustainability of symbiosis	$\delta \geq \delta^* = \frac{u^D - u^C}{u^D - u^P}$	Patience sustains symbiosis
Exit ceiling	$\tau^{\text{exit}} = 1 - \sqrt{2(\bar{a}_A + k)}$	Exit costs bound extraction
Nash bargaining with voice	$\partial \tau^{\text{NB}} / \partial \theta > 0$	Voice bounds extraction
Predation principle	$K^*(r) = \frac{1-r}{2}$	Patient predators conserve hosts
Maximum sustainable injustice	Finite $\bar{G}$ exists	Injustice has a ceiling
Rebalancing rule	$\lambda_i \partial u_i / \partial s_i = \lambda_j \partial u_j / \partial s_j$	Justice as weighted marginal equalization
Institutional stabilization of cooperation	$\pi > c + t$ and $w > c + p$	Conditions for cooperation as an ESS
Collapse	Collapse in finite time absent justice	Necessity of justice mechanisms
Oscillation	Eigenvalues $\pm i\sqrt{bc}$	Dynastic cycle as limit cycle

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